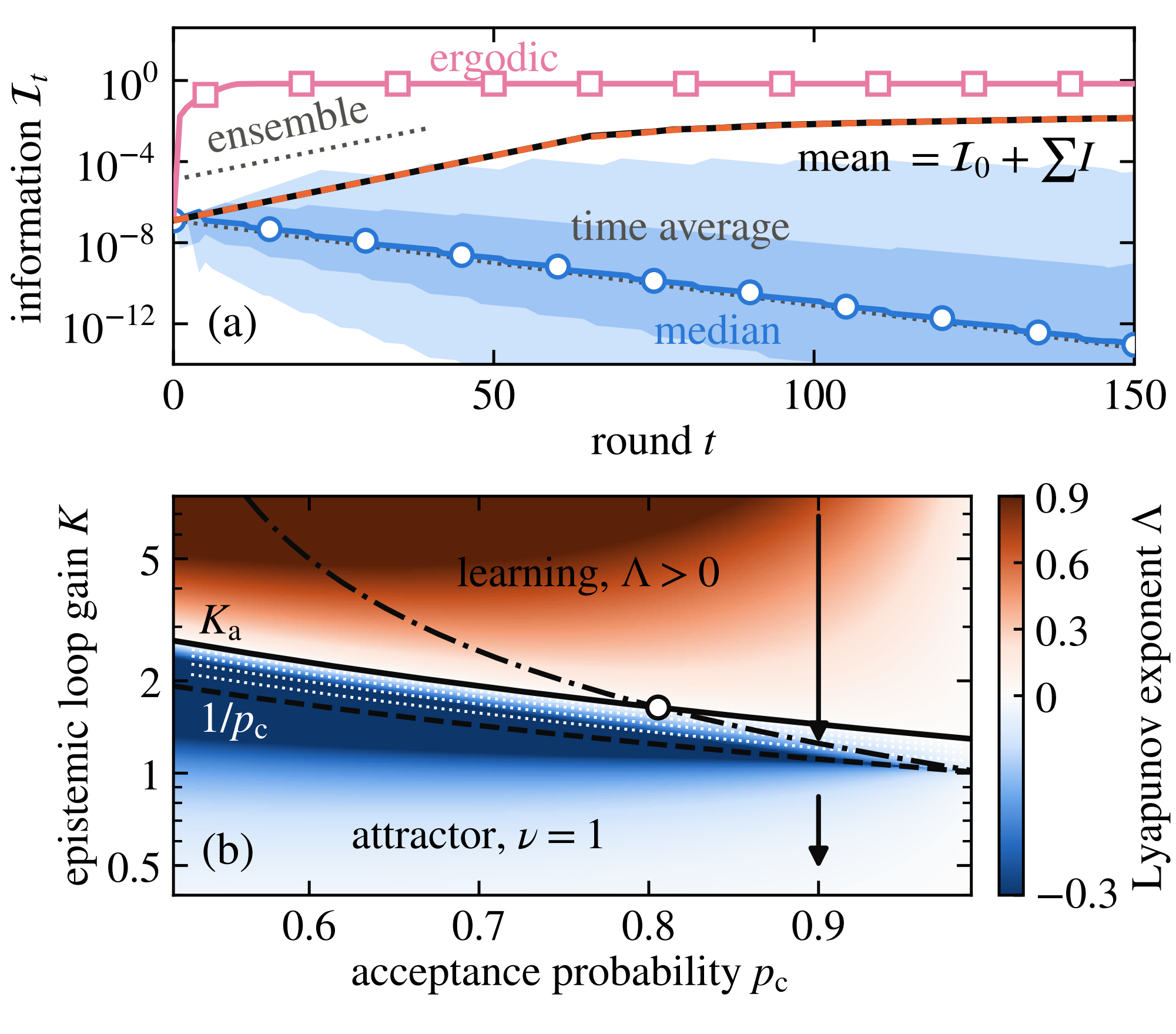

ergodic
ensemble
mean $= \mathcal{I}_0 + \sum I$
time average
median
(a)
information $\mathcal{I}_t$
10^0
10^{-4}
10^{-8}
10^{-12}
0
50
100
150
round t
learning, $\Lambda > 0$
K_a
$1/p_c$
attractor, $\nu = 1$
(b)
epistemic loop gain K
5
2
1
0.5
0.6
0.7
0.8
0.9
acceptance probability p_c
0.9
0.6
0.3
0
−0.3
Lyapunov exponent Λ